



BIOTECH CENTRE

KUMAR ORGANIC PRODUCTS LIMITED

USHA KRISHNA TOWER

Plot# 36B, RoadNo.3, Jigani Industrial Area, Anekal Taluk, Bangalore – 560 105.India

Tel: 91-80-27825196/27825376/27825733 Fax: =080-27825439

FINAL STUDY REPORT

STUDY TITLE

Repeated challenge test

PROTOCOL

ISO 11930

Test Organisms:

[Escherichia coli (ATCC 8739)

Staphylococcus aureus (ATCC 6538)

Candida albicans (ATCC 10231)

Aspergillus niger (ATCC 16404)]

PRODUCT IDENTITY

FD/SK/468/058-A

FD/SK/468/058-B

FD/SK/468/060-A

FD/SK/468/060-B

AUTHORS

Maya Paul, PhD

M.Duraisamy, MSc

STUDY COMPLETION DATE

MAY 26, 2015

PERFORMING LABORATORY

KOPL Biotech Centre- QC Division

Plot# 36B, , Jigani Industrial Area, Anekal Taluk, Bangalore – 560 105.India

REPORT REFERENCE NO:

KBC/DS/484/05



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Abstract

The study was required by KOPL Research Centre (KRC), Kumar Organic Products Ltd. Bangalore to study the Repeated challenge test study of a blend of Kopcerin and Phenoxyethanol in shampoo and cream formulations. The study was performed according to the modified protocol ISO 11930 using the standard test organisms *Escherichia coli* (ATCC 8739), *Pseudomonas aeruginosa* (ATCC 9027), *Staphylococcus aureus* (ATCC 6538), *Candida albicans* (ATCC 10231) and *Aspergillus niger* (ATCC 16404). According to the study the formulations were repeatedly challenged with organisms and it was found that the preservative is competent enough to withstand five cycles of inoculation.



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1.0. STUDY PERSONNEL

1.1. Study Director: Maya Paul, Ph.D

1.2. Professional Personnel involved:

Co-Investigator: Duraisamy M., MSc

2.0. GENERAL STUDY INFORMATION

2.1. Study Title: Repeat Challenge Test

2.2. Report Number: KBC/DS/484/05

2.3. Protocol Number: ISO 11930 Modified

2.4. Sponsor: KRC, Kumar Organic Products Ltd.

2.5. Test Facility: KOPL Biotech Centre

3.0. TEST SUBSTANCE IDENTITY

3.1. Test Substance Name: Kopcerin Phenoxyethanol Blend

3.2. Sample Details:

FD/SK/468/058-A
FD/SK/468/058-B
FD/SK/468/060-A
FD/SK/468/060-B

3.3. Test Substance Characterization:

The test substance characterization as to content, stability, etc. is the responsibility of the Sponsor

4.0. STUDY DATES

4.1. Date Sample Received: March 19, 2015

4.2. Study Initiation Date: April 13, 2015

4.3. Experimental start Date: April 16, 2015

4.4. Experimental End Date: May 25, 2015

4.5. Study Completion Date: May 26, 2015



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5.0. OBJECTIVE

The objective of the study was to analyze the preservative efficacy of the Kopcerin Phenoxyethanol blend in shampoo and cream formulations.

6.0. SUMMARY OF RESULTS

6.2. Test Organisms: *Escherichia coli* (ATCC 8739)
Staphylococcus aureus (ATCC 6538)
Candida albicans (ATCC 10231)
Aspergillus niger (ATCC 16404)

6.3. Exposure Time: Challenge of 7, 14, 21, 28, 35 and 42 days

6.4. Exposure Temperature 22.5 ± 2.5C

6.5. Efficacy Result: All the test samples meet the study acceptance criteria of minimal log reduction values required with an acceptance range of +/-0.5 lg upto 5 inoculation cycles

7.0. STUDY MATERIALS

7.1. Test/Growth Media: Soybean Casein Digest Agar, Sabouraud Dextrose Agar

7.2. Reagents: Diluent, Eugonic LT 100 Broth

7.3 Carriers: NA

8.0. TEST METHOD

8.1. Preparation of test material

The test material is obtained from the KRC, KOPL.. Neat samples are used for the test

8.2. Preparation of the test organisms

The selective growth media were inoculated using a stock culture of each organism. Suspensions were prepared with organisms in log phase of growth and diluted against MC Farland standard No.0.5, to obtain final inoculums of 10^8 cfu/ml of bacterial suspension and 10^7 cfu/ml of yeast and fungi.



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8.3. Efficacy Test Procedure

A mixture of bacteria, yeast and moulds is inoculated 6 times (once a week) into the test material with the goal of monitoring the repeated reduction of the germs for this period. The inoculum contains pathogenic microorganisms and germs which are well-known for product spoilage. All species are cultivated separately and mixed directly before the addition, to ensure a constant composition and germ count of the inoculum. Total germ count is approximately 10^7 - 10^8 cfu/ml.

8.4. Exposure Conditions

The containers were stored at $22.5 \pm 2.5^\circ\text{C}$ for the whole duration of the test.

8.5. Test system recovery

After controlled incubation of bacteria at $32.5 \pm 2.5^\circ\text{C}$ for 48 to 72 hrs and yeast and mold at $22.5 \pm 2.5^\circ\text{C}$ for 3 to 5 days, the plates were observed for the growth. The numbers of viable colonies were determined

8.5. Recovery of survivors

After the incubation period of 48-72 hrs for Bacteria, 3 to 5 days for yeast & fungi the efficacy of the preservative can be observed by the presence/absence of colonies.

9.0. STUDY CONTROLS

9.1. Purity Control

A “streak plate for isolation” was performed on the organism culture and following incubation examined in order to confirm the presence of a pure culture. The acceptance criterion for this study control is a pure culture demonstrating colony morphology typical of the test organism.

9.2. Carrier sterility Control

NA

9.3. Media Sterility control

A representative sample of uninoculated media was incubated and visually examined. The acceptance criterion for this study control is lack of growth.



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10.0. STUDY ACCEPTANCE CRITERIA

Evaluation Criteria for the Preservative Efficacy Test

Minimal log reduction values required with an acceptable range of +/- 0.5 lg

	Bacteria			C. albicans			A.niger	
	T7	T14	T28	T7	T14	T28	T14	T28
Criteria A	$\geq 3 \log$	NI ⁰	NI	$\geq 1 \log$	NI	NI	NI	$\geq 1 \log$
a. The above criteria represent the protection capacities of a formulation with regard to microorganisms.								
b. NI: No increase in the microbial population relative to preceding time								

Criteria A: General case with no additional factors considered

As per previous reports, a cosmetic product without growth of microorganisms after 6 inoculation cycles can be considered microbiologically stable for 30 months, which is recommended for cosmetic products.

Criteria B is fulfilled if the sample showed slight microbial growth (+) during the 6 inoculation cycles. In these instances, microbiological stability may be improved significantly by employing control factors not related to the formulation. For example, a protective packaging, such as pump, provides a higher level of protection against consumer contamination than a jar. Following strict Good Manufacturing Practice (GMP) can protect against contamination occurring during the manufacturing process.

11.0. PROTOCOL CHANGES

11.1. Protocol amendments

No protocol amendments were required for this study.

11.2. Protocol Deviations

No protocol deviations occurred during this study.

12.0. DATA ANALYSIS

12.1. Calculations

A. Determination of the initial numbers of microorganisms N and N₀

- a) Calculate N, the number of microorganisms present in the calibrated suspension in cfu/ml using the equation

$$N = C/V \times d$$

where

C is the mean number of colonies counted in duplicate over the plates

V is the volume of inoculum applied to each dish, in milliliter

d is the dilution factor of the counted dilution

N shall be between 1×10^7 and 1×10^8 cfu/ml for bacteria and between 1×10^6 cfu/ml and 1×10^7 cfu/ml for C.albicans and A.niger



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b) Determine N_0 the number of microorganisms inoculated in the formulation at time T_0 by calculation

$$N_0 = N/100$$

N_0 shall be between 1×10^5 cfu/ml or g and 1×10^6 cfu/ml or g for the bacteria and between 1×10^4 cfu/ml and 1×10^5 cfu/ml or g for *C.albicans* and *A. niger*

B. Enumeration of the microorganisms at each sampling time (N_x)

Calculate N_x the number of surviving microorganisms in the contaminated formulation in cfu/ml or g at each sampling time T_x (T_7 , T_{14} and T_{28}) using the equation

$$N_x = C / V \times d$$

where

C is the mean number of colonies counted in duplicate over the plates

V is the volume of inoculum applied to each dish, in milliliter

d is the dilution factor of the counted dilution

C. Reduction in microbial counts

Calculate the rate of reduction $R_x = \lg N_0 - \lg N_x$

D. % Reduction

Calculate % reduction by calculation

$$\lg N_0 - \lg N_x / \lg N_0 \times 100$$

Cfu/ml = No: of Colonies / Vol. of inoculum X Dilution factor

Log 10 Reduction = Log10 Control cfu/ml - Log10 Test cfu/ml

% Reduction = Log10 Control cfu/ml - Log10 Test cfu/ml / Log10 Control cfu/ml x 100

12.2. Statistical Analysis

None used.

13.0. STUDY RECORD RETENTION

All of the original raw data developed exclusively for this study shall be archived at KOPL Biotech Centre, Plot# 36B, Jigani Industrial Area, Anekal Taluk, Bangalore – 560 105.India

These original data include, but are not limited to, the following:

1. Certified copy of final study report.
2. Any protocol amendments/deviation notifications.
3. All handwritten raw data for control and test substances including, but not limited to notebooks, data forms and calculations.
4. All measured data used in formulating the final report
5. Study specific correspondence relating to interpretation and evaluation of data, other than those documents contained in the final study report



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14.0. EXPERIMENTAL RESULTS

Organism	Bacterial Count	Initial Count '0' hr	'0" Cycle Inoculation Results			
			FD/SK/468/058-A	FD/SK/468/058-B	FD/SK/468/060-A	FD/SK/468/060-B
Mixture culture <i>E.coli</i> <i>S. aureus</i> <i>C. albicans</i> <i>A. niger</i>	cfu/ml	1.20×10^6	<1	TNTC	<1	TNTC
	Log ₁₀ value	6.0791	0	NA	0	NA
	Log Reduction	NA	6.0791	0	6.0791	0
	% Reduction	NA	99.99%	0	99.99%	0
Organism	Bacterial Count	Initial Count '0' hr	'1st" Cycle Inoculation Results			
			FD/SK/468/058-A	FD/SK/468/058-B	FD/SK/468/060-A	FD/SK/468/060-B
Mixture culture <i>E.coli</i> <i>S. aureus</i> <i>C. albicans</i> <i>A. niger</i>	cfu/ml	1.20×10^6	<1	TNTC	<1	TNTC
	Log ₁₀ value	6.0791	0	NA	0	NA
	Log Reduction	NA	6.0791	0	6.0791	0
	% Reduction	NA	99.99%	0	99.99%	0
Organism	Bacterial Count	Initial Count '0' hr	'2nd" Cycle Inoculation Results			
			FD/SK/468/058-A	FD/SK/468/058-B	FD/SK/468/060-A	FD/SK/468/060-B
Mixture culture <i>E.coli</i> <i>S. aureus</i> <i>C. albicans</i> <i>A. niger</i>	cfu/ml	1.20×10^6	<1	TNTC	<1	TNTC
	Log ₁₀ value	6.0791	0	NA	0	NA
	Log Reduction	NA	6.0791	0	6.0791	0
	% Reduction	NA	99.99%	0	99.99%	0
Organism	Bacterial Count	Initial Count '0' hr	'3rd" Cycle Inoculation Results			
			FD/SK/468/058-A	FD/SK/468/058-B	FD/SK/468/060-A	FD/SK/468/060-B
Mixture culture <i>E.coli</i> <i>S. aureus</i> <i>C. albicans</i> <i>A. niger</i>	cfu/ml	1.20×10^6	10cfu/ml	TNTC	<1	TNTC
	Log ₁₀ value	6.0791	1.0000	NA	0	NA
	Log Reduction	NA	5.0791	0	6.0791	0
	% Reduction	NA	83.55%	0	99.99%	0



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Organism	Bacterial Count	Initial Count '0' hr	'4th" Cycle Inoculation Results			
			FD/SK/468/ 058-A	FD/SK/468/ 058-B	FD/SK/468/ 060-A	FD/SK/468/ 060-B
Mixture culture <i>E.coli</i> <i>S. aureus</i> <i>C. albicans</i> <i>A. niger</i>	cfu/ml	1.20×10^6	10cfu/ml	TNTC	<1	TNTC
	Log ₁₀ value	6.0791	1.0000	NA	0	NA
	Log Reduction	NA	5.0791	0	6.0791	0
	% Reduction	NA	83.55%	0	99.99%	0
Organism	Bacterial Count	Initial Count '0' hr	'5th" Cycle Inoculation Results			
			FD/SK/468/ 058-A	FD/SK/468/ 058-B	FD/SK/468/ 060-A	FD/SK/468/ 060-B
Mixture culture <i>E.coli</i> <i>S. aureus</i> <i>C. albicans</i> <i>A. niger</i>	cfu/ml	1.20×10^6	10cfu/ml	TNTC	<1	TNTC
	Log ₁₀ value	6.0791	1.0000	NA	0	NA
	Log Reduction	NA	5.0791	0	6.0791	0
	% Reduction	NA	83.55%	0	99.99%	0
Organism	Bacterial Count	Initial Count '0' hr	'6 th" Cycle Inoculation Results			
			FD/SK/468/ 058-A	FD/SK/468/ 058-B	FD/SK/468/ 060-A	FD/SK/468/ 060-B
Mixture culture <i>E.coli</i> <i>S. aureus</i> <i>C. albicans</i> <i>A. niger</i>	cfu/ml	1.20×10^6	60 cfu/ml	TNTC	<1	TNTC
	Log ₁₀ value	6.0791	1.7781	NA	0	NA
	Log Reduction	NA	4.3010	0	6.0791	0
	% Reduction	NA	70.75%	0	99.99%	0



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15.0. STUDY CONCLUSION

The study employing modified protocol ISO 11930 using the standard test organisms *Escherichia coli* (ATCC 8739), *Pseudomonas aeruginosa* (ATCC 9027), *Staphylococcus aureus* (ATCC 6538), *Candida albicans* (ATCC 10231) and *Aspergillus niger* (ATCC 16404), it has been proved that the shampoo and cream formulations preserved with the blend of Kopcerin and Phenoxyethanol are able to withstand five cycles of inoculation without any issues of microbial growth. The placebo without the preservative blend is observed to be harbored with microorganisms confirming the authenticity rationality of the study.

